

Forward Problem

parametri iniziali:

- $\mathbf{x}_0, \mathbf{v}_0$: stato iniziale
- $\mathbf{k}, \alpha$: params materiale
- $\mathbf{g}$: gravità
- $\mathbf{f}_{\text{ext}}$: forze esterne

traiettoria

$$\begin{bmatrix} \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{\text{end}} \\ \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{\text{end}} \end{bmatrix}$$

θ_{init}



$\mathbf{F}$



$\bar{\mathbf{x}}, \bar{\mathbf{v}}$

simulatore

$$\mathbf{F}_t(\mathbf{x}_t, \mathbf{v}_t, \theta_t) = (\mathbf{x}_{t+1}, \mathbf{v}_{t+1})$$

Inverse Problem

traiettoria parziale

es: $\{ \mathbf{x}_{31}, \mathbf{x}_{47}, \mathbf{x}_{\text{end}} \}$

parametri

tali che:

$$\mathbf{F}(\boldsymbol{\theta}) \approx \tilde{\mathbf{x}}, \tilde{\mathbf{v}}$$

$\tilde{\mathbf{x}}, \tilde{\mathbf{v}}$



Q

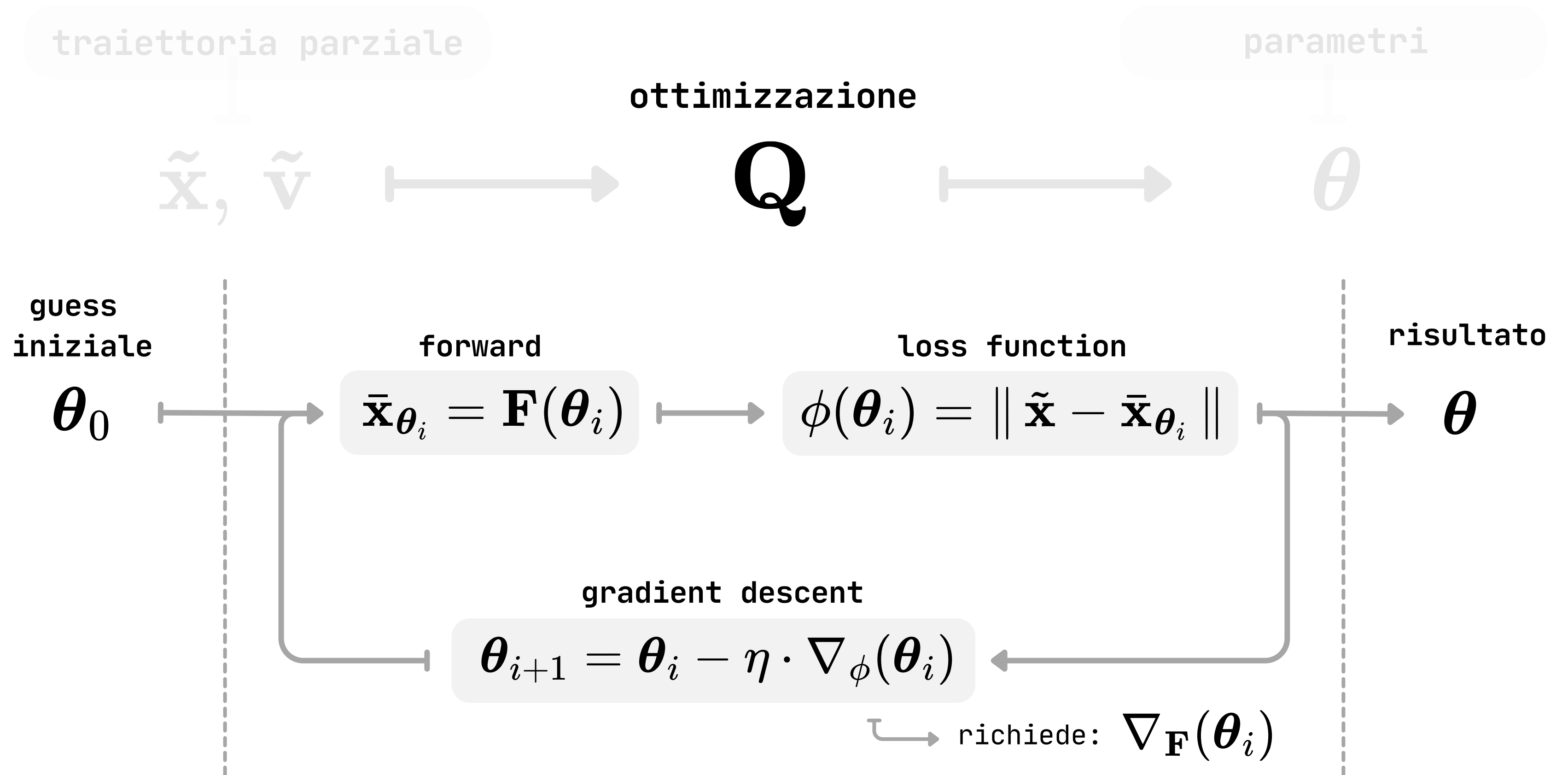


$\boldsymbol{\theta}$

problema di ottimizzazione

cerco i parametri che, usati come input del simulatore forward,
producono la traiettoria più vicina a quella osservata

Inverse Problem



Integratore

metodo esplicito

$$\mathbf{v}_{t+1} = \mathbf{v}_t + h \mathbf{M}^{-1} \mathbf{f}(\mathbf{x}_t, \mathbf{v}_t)$$

$$\mathbf{x}_{t+1} = \mathbf{x}_t + h \mathbf{v}_{t+1}$$

le forze dipendono dallo
stato corrente:

- calcolo diretto
- instabile per forze rigide

metodo implicito

$$\mathbf{v}_{t+1} = \mathbf{v}_t + h \mathbf{M}^{-1} \mathbf{f}(\mathbf{x}_{t+1}, \mathbf{v}_{t+1})$$

$$\mathbf{x}_{t+1} = \mathbf{x}_t + h \mathbf{v}_{t+1}$$

le forze dipendono dallo
stato futuro:

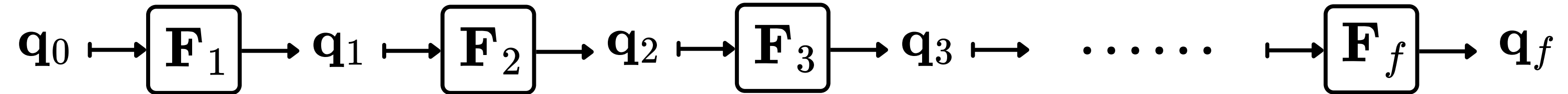
- stabile
- richiede risolvere un sistema
nonlineare

$$\mathbf{q}_{t+1} = \mathbf{F}(\mathbf{q}_t, \mathbf{q}_{t+1}, \boldsymbol{\theta})$$

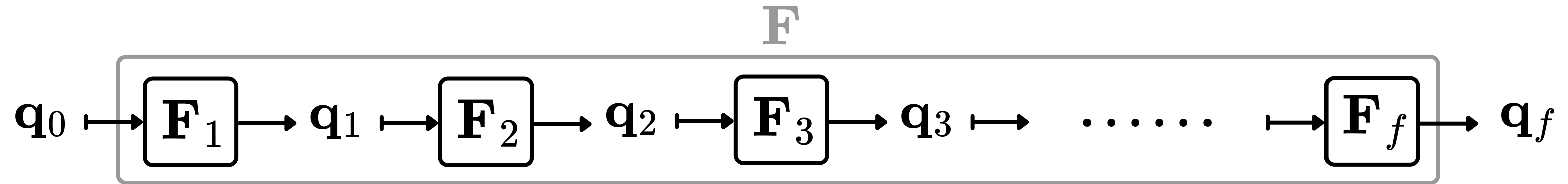


- Newton-Raphson
- Quasi-Newton
- Baraff-Witkin
- Projective Dynamics
- XPBD

Esempio: ottimizzare su q_0



Esempio: ottimizzare su $\mathbf{q}_0$



A fine simulazione mi voglio avvicinare il più possibile allo stato obiettivo:

$\mathbf{q}_{\text{goal}}$

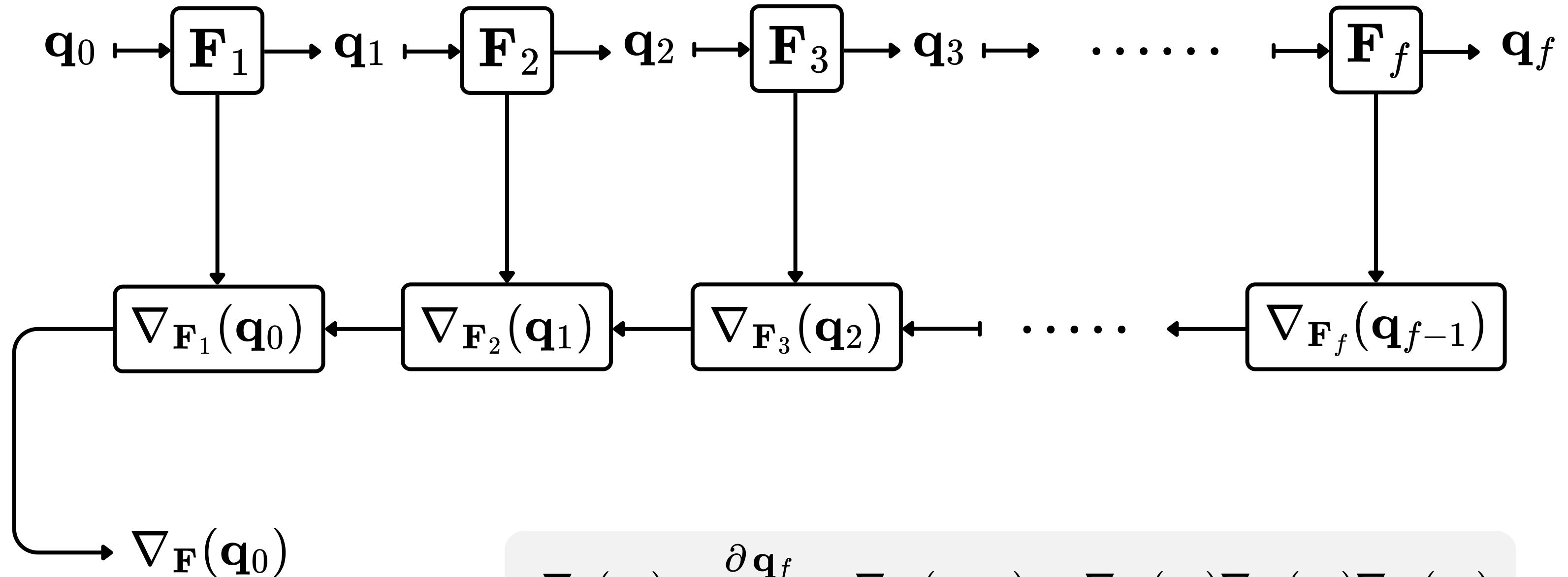
Ottimizzo sullo stato iniziale della simulazione:

$\mathbf{q}_0$

Cerco lo stato iniziale tale che all'ultimo istante di simulazione sono il più vicino possibile allo stato obiettivo:

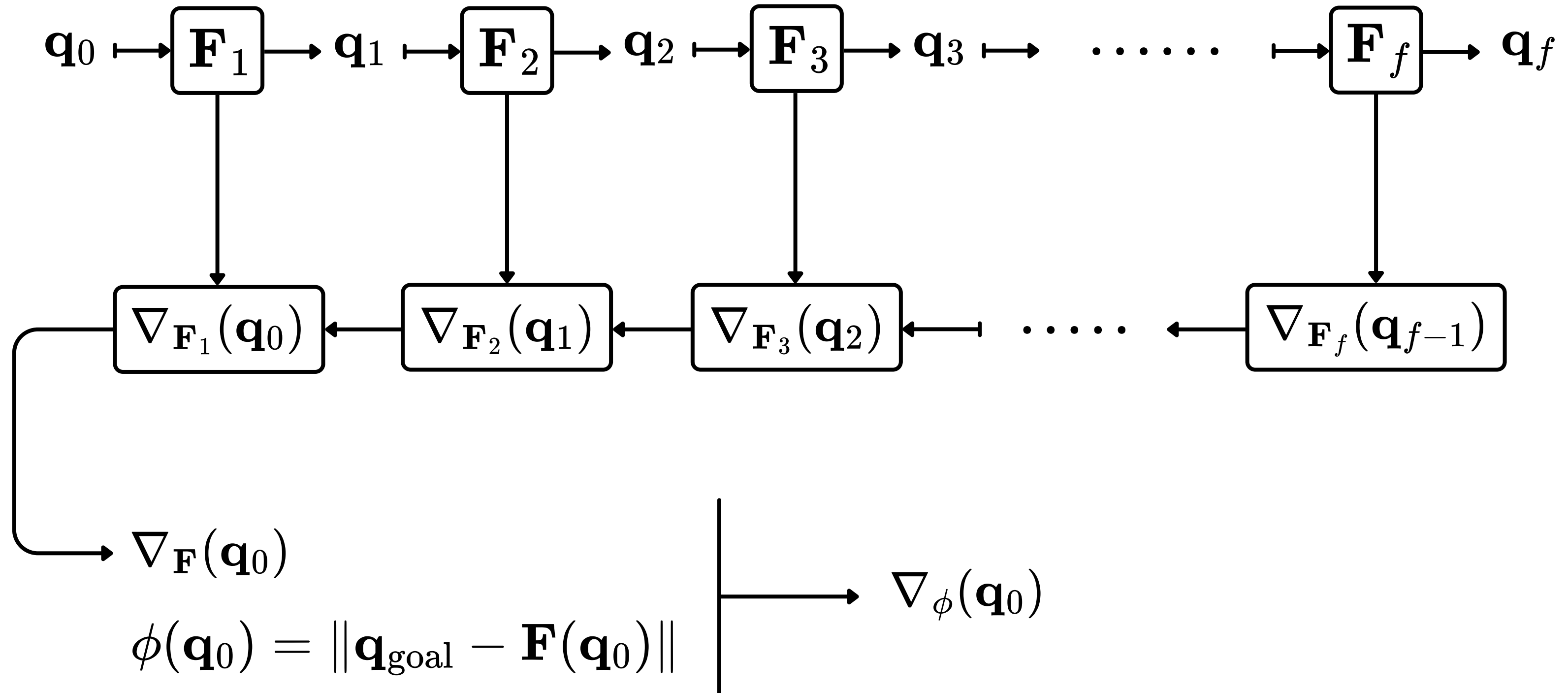
$$\phi(\mathbf{q}_0) = \|\mathbf{q}_{\text{goal}} - \mathbf{F}(\mathbf{q}_0)\|$$

Esempio: ottimizzare su $\mathbf{q}_0$

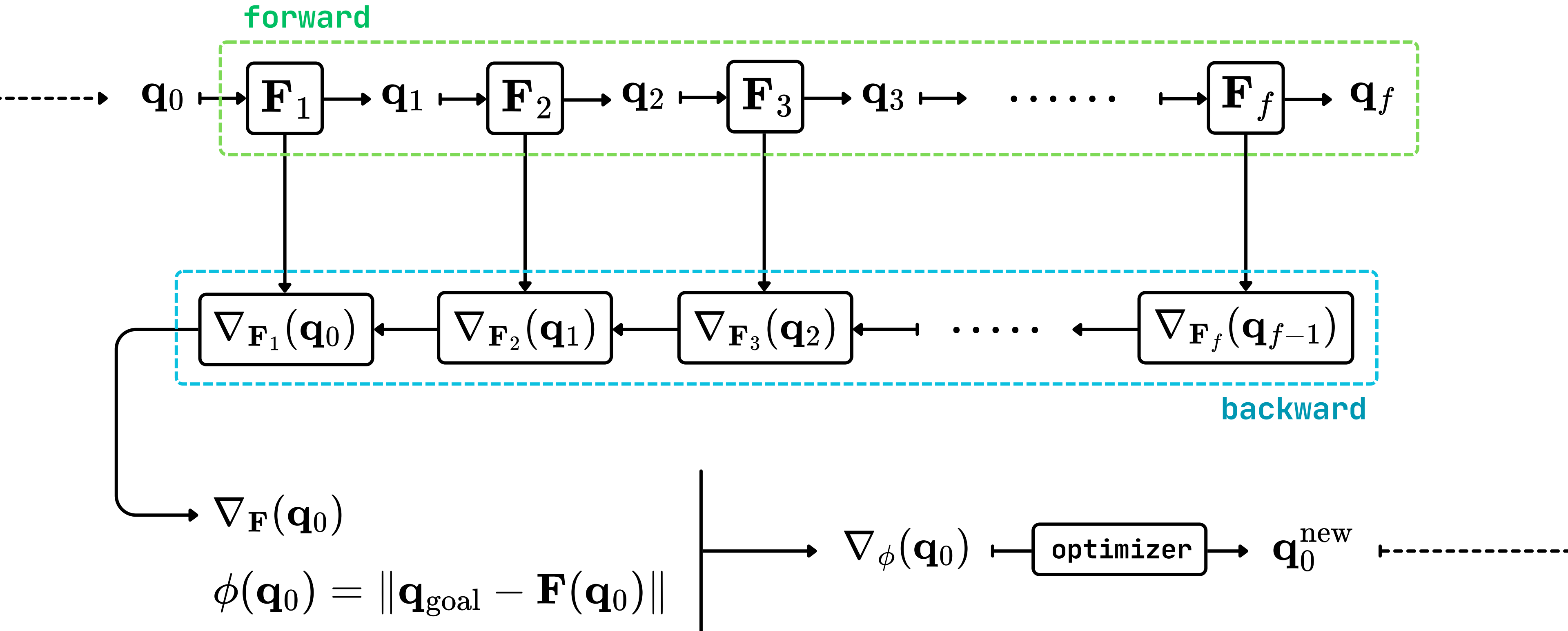


$$\nabla_{\mathbf{F}}(\mathbf{q}_0) = \frac{\partial \mathbf{q}_f}{\partial \mathbf{q}_0} = \nabla_{\mathbf{F}_f}(\mathbf{q}_{f-1}) \cdots \nabla_{\mathbf{F}_3}(\mathbf{q}_2) \nabla_{\mathbf{F}_2}(\mathbf{q}_1) \nabla_{\mathbf{F}_1}(\mathbf{q}_0)$$

Esempio: ottimizzare su $\mathbf{q}_0$



Esempio: ottimizzare su $\mathbf{q}_0$



XPBD

Algorithm XPBD Time Step with 1 Iteration

```
1: function  $\mathbf{f}_{\text{XPBD}}(\mathbf{q}_t = [\mathbf{x}_t, \mathbf{v}_t], \tilde{\mathbf{A}})$ 
2:    $\mathbf{x} \leftarrow \tilde{\mathbf{x}}$ 
3:   for  $i$  constraints do
4:      $\Delta\lambda_i \leftarrow \frac{-\mathbf{C}_i(\mathbf{x})}{\mathbf{J}_i\mathbf{M}^{-1}\mathbf{J}_i^\top + \tilde{\boldsymbol{\alpha}}_i}$ 
5:      $\Delta\mathbf{x}_i \leftarrow \mathbf{M}^{-1}\mathbf{J}_i \Delta\lambda_i$ 
6:    $\Delta\mathbf{x} \leftarrow \Delta\mathbf{x}_1 + \Delta\mathbf{x}_2 + \cdots + \Delta\mathbf{x}_m$ 
7:    $\mathbf{x}_{t+1} \leftarrow \tilde{\mathbf{x}} + \Delta\mathbf{x}$ 
8:    $\mathbf{v}_{t+1} \leftarrow (\mathbf{x}_{t+1} - \mathbf{x}_t) / h$ 
9: end function
```

XPBD

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```

$$\frac{\partial \mathbf{f}_{\text{XPBD}}}{\partial \mathbf{q}_t} = \begin{bmatrix} \frac{\partial \mathbf{x}_{t+1}}{\partial \mathbf{x}_t} & \frac{\partial \mathbf{x}_{t+1}}{\partial \mathbf{v}_t} \\ \frac{\partial \mathbf{v}_{t+1}}{\partial \mathbf{x}_t} & \frac{\partial \mathbf{v}_{t+1}}{\partial \mathbf{v}_t} \end{bmatrix}$$

$$\frac{\partial \mathbf{x}_{t+1}}{\partial \mathbf{x}_t} = \mathbf{I} + \frac{\partial \Delta\mathbf{x}}{\partial \mathbf{x}_t}$$

$$\frac{\partial \mathbf{x}_{t+1}}{\partial \mathbf{v}_t} = h \left(\mathbf{I} + \frac{\partial \Delta\mathbf{x}}{\partial \mathbf{x}_t} \right)$$

$$\frac{\partial \mathbf{v}_{t+1}}{\partial \mathbf{x}_t} = \frac{1}{h} \frac{\partial \Delta\mathbf{x}}{\partial \mathbf{x}_t}$$

$$\frac{\partial \mathbf{v}_{t+1}}{\partial \mathbf{v}_t} = \mathbf{I} + \frac{\partial \Delta\mathbf{x}}{\partial \mathbf{x}_t}$$
